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1  # Full RF Rectifier Report
2
3  ## 1. Optimization Simulation Setup
4  | Item | Value |
5  |-----|-----|
6  | RF Frequency (Hz) | 920 MHz |
7  | RF Input Amplitude (V) | 0.5 |
8  | Temperature (°C) | 27 |
9  | Supply Voltage VDD (V) | 0 |
10 | Process Corner | tt |
11 | Population Size | 50 |
12 | GA Max Generations | 10 |
13 | DE Max Generations | 20 |
14 | Crossover Probability (pcross) | 0.9 |
15 | Mutation Probability (pmut) | 0.2 |
16 | Fscale | 0.7 |
17 | Crossover Rate (CR) | 0.8 |
18 | Stall Generation Limit | 10 |
19 | Acceptance Tolerance | 1e-12 |
20 | Stop Cost Threshold | acceptTol |
21 | PVT Corners | tt, ss, ff, sf, fs |
22 | VDD Sweep (V) | 0 |
23 | Temperature Sweep (°C) | -40, 0, 27, 70, 85 |
24
25 ---
26
27 ## 2. Best Solution Summary
28 | Parameter | Value |
29 |-----|-----|
30 | Generation | 19 |
31 | Index | 18 |
32 | Total Index | 1018 |
33 | VOUTMAX | 0.522 V |
34 | POUT | 1.20 µW |
35 | PIN | 1.27 µW |
36 | PCE | 94.99 % |
37 | Sensitivity (PIDBM) | -28.98 dBm |
38 | Ripple (ΔV) | 5.19 mV |
39 | Area | 634 µm2 |
40 | FOM1 | 707.91 |
41 | FOM2 | 20514.0 |
42 | FOM3 | 39264.7 |
43
44 ---
45
46 ## 3. Constraint Verification
47 | Metric | Target | Result | Status |
48 |-----|-----|-----|-----|
49 | VOUT | 0.5-1 V | 0.522 V | PASS |
50 | POUT | ≥1 µW | 1.20 µW | PASS |
51 | PCE | 85-95 % | 94.99 % | PASS |
52 | PIDBM | -35 to -15 dBm | -28.98 | PASS |
53 | Ripple | ≤10 mV | 5.19 mV | PASS |
54 | Area | within bounds | 634 µm2 | PASS |
55
56 ---
57
58 ## 4. PVT Summary (Extremes)
59 - VOUT: **0.488 - 0.548 V**
60 - POUT: **1.04 - 1.32 µW**
61 - PCE: **74.57 - 98.04 %**
62 - PIDBM: **-28.99 to -28.25 dBm**
63 - Ripple: **4.33 - 8.81 mV**
64
65 ---
66
67 ## 5. Monte Carlo Summary
68 | Metric | Mean | Sigma |
69 |-----|-----|-----|

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70 | VOUT | 0.522 V | 3.11 mV |
71 | POUT | 1.199  $\mu$ W | 14.5 nW |
72 | PCE | 94.51 % | 1.35 % |
73 | PIDBM | -28.64 dBm | 0.22 dB |
74 | Ripple | 5.23 mV | 0.22 mV |
75
76 ---
77
78 ## 6. Yield Report
79 - Total Samples: 1000
80 - Passing Samples: 691
81 - **Yield: 69.10%**
82
83 ---
84
85 ## 7. RF Performance Metrics
86 - Dynamic Range (DR): **23.05 dB**
87 - PCE Peak: **94.99 % @ -28.98 dBm**
88 - Sensitivity (@1 V): **-18.58 dBm**
89
90 ---
91
92 ## 8. Impedance Matching (X-Ray)
93 - Input Impedance (pre-match): **1.01 - j149  $\Omega$ **
94 - S11 (before): **-0.036 dB**
95 - S11 (after): **-26.08 dB**
96 - Matched Impedance: ** $\approx$  50  $\Omega$ **
97
98 ---
99
100 ## 9. Runtime Summary
101 - Total Runtime: **3.82 hours**
102
```